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BI Practical 7

Problem Statement: Data Visualization from Extraction Transformation and Loading (ETL) Process.

Step 1: Check Data (Extraction)

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("D:/College/BE/Dataset/Iris.csv")

print(df.head())
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Step 2: Transform

Check missing values

```
print(df.isnull().sum())
```

```
SepalLengthCm    0
SepalWidthCm     0
PetalLengthCm    0
PetalWidthCm     0
Species          0
dtype: int64
```

Create new column

```
df['PetalArea'] = df['PetalLengthCm'] * df['PetalWidthCm']
```

Convert species names

```
df['Species'] = df['Species'].str.replace('Iris-', '')
```

```
df.head()
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	PetalArea
0	5.1	3.5	1.4	0.2	setosa	0.28
1	4.9	3.0	1.4	0.2	setosa	0.28
2	4.7	3.2	1.3	0.2	setosa	0.26
3	4.6	3.1	1.5	0.2	setosa	0.30
4	5.0	3.6	1.4	0.2	setosa	0.28